

浙江省 2026 年初中学业水平模拟考试

数 学 参 考 答 案

一、选择题

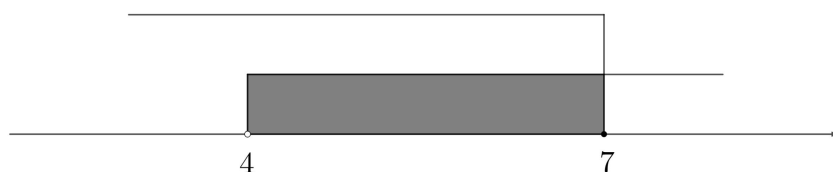
1	2	3	4	5	6	7	8	9	10
A	D	B	D	D	C	B	C	A	A

二、填空题

11	12	13	14	15	16
11	45°	$\frac{48-25\sqrt{3}}{11}$	$\sqrt{17}-1$	$m < 2$ 或 $2 < m \leq \frac{16}{3}$ 或 $m \geq 8$	$2\sqrt{21}$

三、解答题

17. $(x+4)(x-4)$



18. $4 < x \leq 7$

19.(1) $l_1 = 4t(0 \leq t \leq 900)$, $l_2 = 3t - 360(120 \leq t \leq 900)$	(2) 3 次
--	---------

20.(1) 275	(2) -62, 17.2	(3) $y = \begin{cases} 0, & 0 \leq x < 0.5 \\ 1, & x \geq 0.5 \end{cases}$
------------	---------------	--

21.(1) 5	(2) 9
(3) 错误, ① “$/-7x+70/=2$” 应为 “$/-7x+70/=1$”. ② “$/-7x/=1$” 应为 “$/-7x/=9$”. 规避: 优化代入方式, 使代数表达式尽可能保持非负, 系数为正. 如何: ①先审视工具的前提, 理解透彻概念. ②抓住问题特殊性, 调整表达形式, 注重细节, 细节决定成败. ③最后才应用简化策略.	

22.(1) 略	(2) $\frac{\sqrt{73}}{2}$ 或 $\frac{\sqrt{53}}{2}$	(3) 53, $\frac{gcd(p_1q_2, p_2q_1)}{q_1q_2}$
----------	---	--

23.(1) $y = 3x^2 - 15x + 12$	(2) $y' = 3x + p - 9, p > -6$	(3) 略
------------------------------	-------------------------------	-------

24.(1) 略	(2) $\frac{\sqrt{10}}{5}$	(3) 略
----------	---------------------------	-------

精准解析

一、选择题

1. 负数是小于 0 的数, 在选项中只有 -7 小于 0.

2. 总共有 9 张卡片, 其中写有“学”字的卡片有 4 张, 抽到“学”字的概率为 $\frac{4}{9}$.

3. 科学记数法表示的形式为 $a \times 10^n$, 其中 $1 \leq |a| < 10$, n 是整数. 将 0.0000025 的小数点向右移动 7 位得到 2.5, 所以 $n = -7$. 因此, 0.0000025 用科学记数法表示为 2.5×10^{-7} .

4. 解: 由图知 AD 为 $\angle BAD$ 的角平分线, $AB = AC$.

$\because AB = AC, AD$ 平分 $\angle BAD$,

$\therefore AD \perp BC, D$ 为 BC 中点.

而 AB 可以不等于 BC .

5. ① $= 2^2 \times 2 = 2^3$, 正确.

② $= (x^3)^3 = x^{3 \times 3} = x^9$, 正确.

③ $= x^2 + y^2 + 2xy$, 不正确.

④ $\begin{cases} -k^3 \geq 0 \\ k \neq 0 \end{cases}$, 解得 $k < 0$.

$\therefore \frac{\sqrt{-k^3}}{k} = \frac{\sqrt{k^2 \cdot (-k)}}{k} = \frac{-k\sqrt{-k}}{k} = -\sqrt{-k} \neq 0$.

$\therefore m = n$, 正确.

6. 解: 过点 C 作 $CE \perp x$ 轴于点 E , 过点 C' 作 $C'F \perp x$ 轴于点 F .

$\because CE \perp x$ 轴, $C'F \perp x$ 轴,

$\therefore \angle CEO = \angle C'FO = 90^\circ$.

$\because \triangle ABC \sim \triangle A'B'C'$,

$\therefore \frac{S_{\triangle ABC}}{S_{\triangle A'B'C'}} = \frac{1}{4\sqrt{2}} = \left(\frac{OC}{OC'}\right)^2$.

在 $\triangle COE$ 与 $\triangle C'OF$ 中,

$\begin{cases} \angle COE = \angle C'OF \\ \angle CEO = \angle C'FO = 90^\circ \end{cases}$

$\therefore \triangle COE \sim \triangle C'OF$.

设 C' 在 $y = \frac{m}{x}$ 上.

$\therefore \left(\frac{OC}{OC'}\right)^2 = \frac{S_{\triangle COE}}{S_{\triangle C'OF}} = \frac{\frac{1}{2}OE \cdot EC}{\frac{1}{2}OF \cdot C'F} = \frac{8}{m}$.

$\therefore m = 32\sqrt{2}$.

$\because$ 由对称性知 C' 关于 y 轴的对称点在 $y = -\frac{32\sqrt{2}}{x}$ 上.

$\therefore k = 32\sqrt{2}$.

7. 第一步: 求水平距离 (侧面三角形)

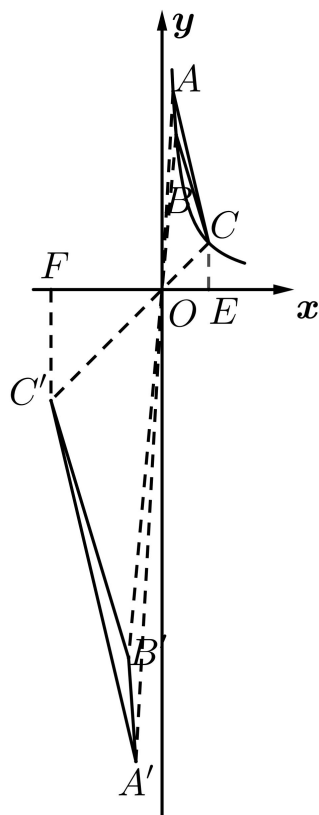
设无人机位置为 P , 其在地面的投影为 P' .

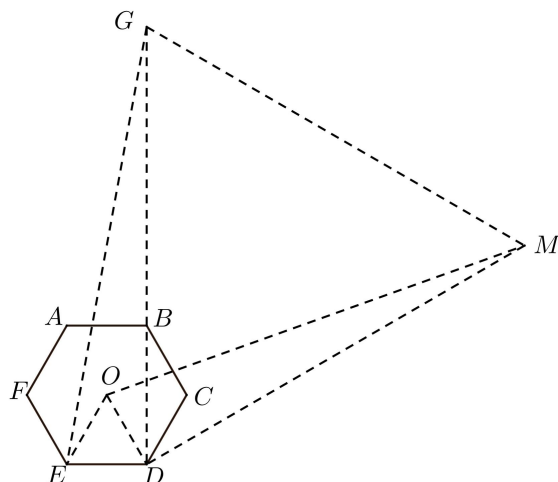
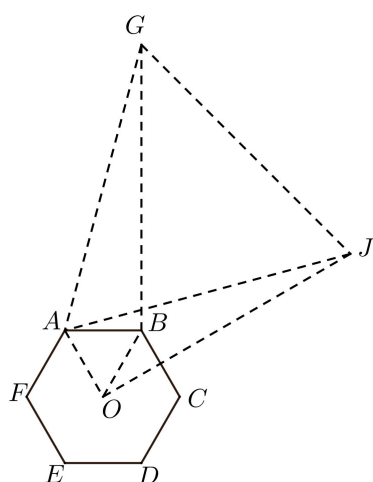
在 $\text{Rt}\triangle PP'O$ 中, 已知高度 $PP' = 120$, 俯角 $\angle POP' = \alpha$.

由此求出水平总距离 (即底边 OP'):

$$OP' = \frac{120}{\tan \alpha}$$

第二步: 求向南距离 (俯视直角三角形)





10.解: 在 AB 上取点 F, 使 $\angle BFC = 45^\circ$.

$$\because \angle AEB = 135^\circ$$

$$\therefore \angle BEC = \angle BFC = 45^\circ$$

$$\therefore \angle AEB = \angle AFC = 135^\circ$$

在 $\triangle AEB$ 与 $\triangle AFC$ 中

$$\therefore \begin{cases} \angle BAE = \angle FAC \\ \angle AEB = \angle AFC \end{cases}$$

$$\therefore \triangle AEB \sim \triangle AFC$$

$$\therefore \frac{AE}{AF} = \frac{AB}{AC}$$

$$\therefore AE \cdot AC = AB \cdot AF = 4$$

连结 CF, 过点 D 作 $DG \perp CF$ 交 CF 于点 G,

过点 C 作 $CH \perp AB$ 交 AB 于点 H

$$\because DG \perp FC, \angle DFC = 45^\circ$$

$$\therefore \angle FGD = 90^\circ, \angle FDG = 45^\circ$$

$$\because \angle ADC = 75^\circ$$

$$\therefore \angle GDC = \angle FDC - \angle FDG = 30^\circ$$

在 $\text{Rt}\triangle DGC$ 中, $\angle DGC = 90^\circ, \angle GDC = 30^\circ$,

设 $GC = k$

$$\text{则 } DG = \sqrt{3}k$$

在 $\text{Rt}\triangle FGD$ 中, $\angle GFD = 45^\circ, \angle FGD = 90^\circ$

$$\therefore FG = DG = \sqrt{3}k, DF = \sqrt{6}k$$

$$\therefore S_{\triangle DFC} = \frac{1}{2} \cdot CF \cdot DG = \frac{1}{2} \cdot DF \cdot CH$$

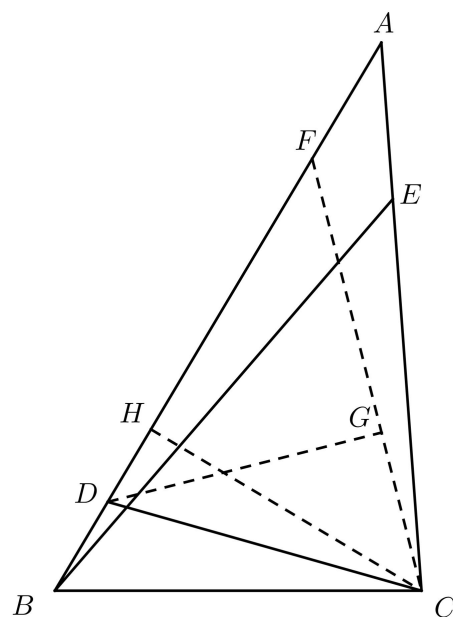
$$\therefore \frac{\sqrt{3}(\sqrt{3}+1)k^2}{2} = \frac{\sqrt{6}k \cdot CH}{2} \quad (1)$$

$$\because S_{\triangle ABC} = \frac{1}{2} \cdot AB \cdot CH = 3 + \sqrt{3} \quad (2)$$

$$\therefore \text{由 } (1)(2) \text{ 知 } k \cdot AB = 2\sqrt{6}$$

$$\therefore DF \cdot AB = \sqrt{6}k \cdot AB = 12$$

$$\therefore AD \cdot AB = (AF + FD) \cdot AB = AF \cdot AB + FD \cdot AB = 4 + 12 = 16.$$



二、填空题

11.解: 原式 $= 15m - 10 - 15m + 21 = -10 + 21 = 11$

12.解: $\because \angle A + \angle B = 3\angle C, \angle A + \angle B + \angle C = 180^\circ$

$$\therefore \angle A + \angle B + \angle C = 4\angle C = 180^\circ$$

$$\therefore \angle C = 45^\circ$$

13.解: 过点A作 $AE \perp BC$ 交BC于点E

过点D作 $DF \perp AE$ 交AE于点F

过点B作 $BG \perp AD$ 交AD于点G

$$\because \angle BAE + \angle EAC = 90^\circ$$

$$\angle EAC + \angle ACE = 90^\circ$$

$$\therefore \angle BAE = \angle ACE$$

在 $Rt \triangle BAE$ 中, 设 $BA = 3$, $\tan \angle BAE = \frac{3}{4}$

$$\therefore BE = \frac{9}{5}, EA = \frac{12}{5}$$

$$\because \angle ADF = \angle BDA - \angle BDF = 30^\circ, DF = BE = \frac{9}{5}$$

$$\therefore \text{在} Rt \triangle ADF \text{中}, AF = \frac{3\sqrt{3}}{5}, DA = \frac{6\sqrt{3}}{5}$$

$$\therefore DB = \frac{12 - 3\sqrt{3}}{5}$$

再将边长放大为原先 10 倍

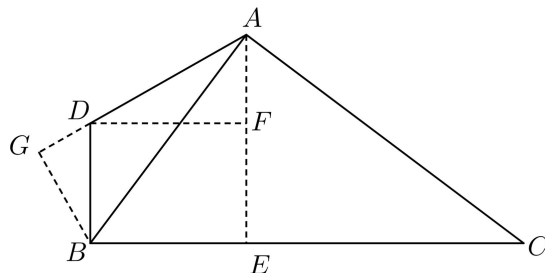
$$\therefore DA = 12\sqrt{3}, DB = 24 - 6\sqrt{3}$$

$$\because \text{在} Rt \triangle DBG \text{中}, \angle GDB = 180^\circ - \angle BDA = 60^\circ$$

$$\therefore GD = 12 - 3\sqrt{3}, GB = 12\sqrt{3} - 9$$

$$\because \text{在} Rt \triangle BGA \text{中}, \tan \angle DAB = \frac{GB}{GA}$$

$$\therefore \tan \angle DAB = \frac{12\sqrt{3} - 9}{12 + 9\sqrt{3}} = \frac{4\sqrt{3} - 3}{4 + 3\sqrt{3}} = \frac{48 - 25\sqrt{3}}{11}$$



14.解: 延长 CD 交 AB 于点 E

设 $\angle BAD = \angle DBC = \alpha$, 则 $\angle DCB = 90^\circ - 2\alpha$

$$\because \angle ABC = 90^\circ$$

$$\therefore \angle CEB = 2\alpha$$

$$\because \angle CEB = \angle EAD + \angle ADE$$

$$\therefore \angle ADE = \angle DAE = \alpha, AE = ED$$

$$\because \angle DBE = \angle EBC - \angle DBC = 90^\circ - \alpha$$

$$\therefore \angle EDB = \angle DBE = 90^\circ - \alpha$$

$$\therefore DE = EB$$

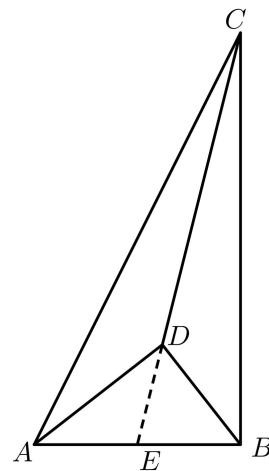
$$\therefore AE = EB$$

$$\because \angle ADB = 180^\circ - \angle DAB - \angle DBA = 90^\circ$$

$$\therefore DE = \frac{1}{2}AB = 1$$

在 $Rt \triangle CBE$ 中, $CE = \sqrt{CB^2 + BE^2} = \sqrt{17}$

$$\therefore DC = CE - DE = \sqrt{17} - 1$$



15.解: $(x - m)(x + 4) + (2x + m)(x - 2) = 4(x + 4)(x - 2)$

化简得 $x^2 + 8x - 32 + 6m = 0$, 其对称轴为直线 $x = -4$.

当此方程仅有负数根时:

$$\textcircled{1} \begin{cases} x \neq 2 \\ x \neq -4 \\ \Delta \geq 0 \\ f(0) > 0 \end{cases}, \text{解得 } \frac{16}{3} < m < 8$$

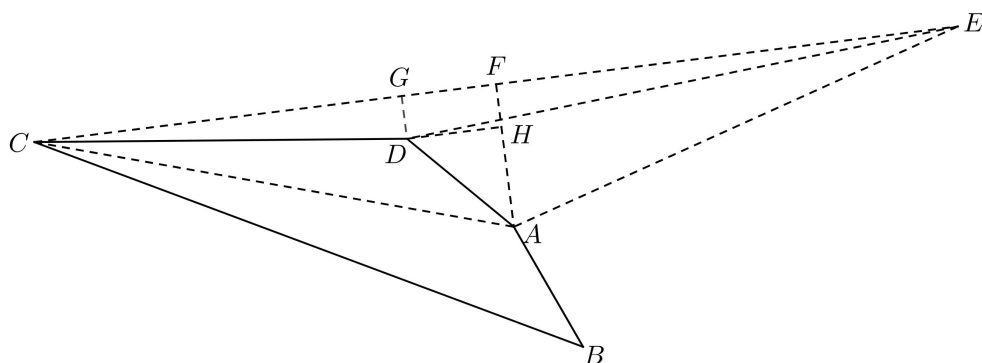
$$\textcircled{2} f(2) = 0, \text{解得 } m = 2$$

此时二次方程有两根,但对于分式方程而言需舍去 $x = 2$,故仅有负根.

$\therefore$ 其为假命题

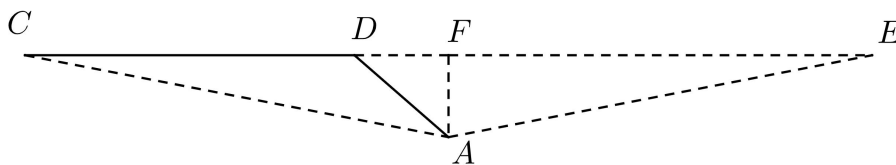
$$\therefore m < 2 \text{ 或 } 2 < m \leq \frac{16}{3} \text{ 或 } m \geq 8$$

16.解:



将 $\triangle BAC$ 旋转至 $\triangle DAE$,
 过点 A 作 $AF \perp CE$ 交 CE 于点 F ,
 过点 D 作 $DG \perp CE$ 交 CE 于点 G ,
 过点 D 作 $DH \perp AF$ 交 AF 于点 H
 $\because \triangle BAC \cong \triangle DAE$
 $\therefore CA = AE$
 $\therefore AF \perp CE$
 $\therefore CF = FE$
 设 $GF = y$, $FE = x$
 $\therefore CG = CF - GF = FE - GF = x - y$
 在 $Rt \triangle CGD$ 中, $GD^2 = CD^2 - CG^2$
 $= 7^2 - (x - y)^2$
 在 $Rt \triangle EGD$ 中, $GD^2 = DE^2 - GE^2$
 $= 11^2 - (x + y)^2$
 $\therefore 7^2 - (x - y)^2 = 11^2 - (x + y)^2$
 $\therefore xy = 18$
 $\therefore S_{ABCD} = S_{CDA} + S_{ADE} = S_{CAE} - S_{CDE}$
 $= \frac{1}{2}CE \cdot AF - \frac{1}{2}CE \cdot GD$
 $= \frac{1}{2}CE \cdot (AF - GD)$

$= \frac{1}{2}CE \cdot HA$
 当 C, D, E 三点共线时, CE 取到最大值,
 即 $2x$ 取到最大值
 $\therefore xy = 18$
 $\therefore y$ 此时取到最小值
 在 $Rt \triangle DHA$ 中, $DA = \sqrt{7}$, $DH = y$
 $HA = \sqrt{DA^2 - DH^2} = \sqrt{7 - y^2}$
 $\therefore HA$ 此时取到最大值
 $\therefore$ 此时 S_{ABCD} 取到最大值
 此时 G, D 共点, F, H 共点, $CE = CD + DE = 18$
 $\therefore CA = AE, AF \perp CE$
 $\therefore CF = FE = \frac{1}{2}CE = 9 \therefore CD = 7$
 $\therefore DF = CF - CD = 2$
 在 $Rt \triangle DFA$ 中,
 $FA = \sqrt{DA^2 - DF^2} = \sqrt{3}$
 在 $Rt \triangle CFA$ 中,
 $CA = \sqrt{FA^2 + CF^2} = 2\sqrt{21}$



三、解答题

17.解: 当我们遇到平方减平方的式子要因式分解时,可以考虑一下平方差公式
 原式 $= (x + 4)(x - 4)$

18.解: $\begin{cases} 3x + 8 \leq 29 & \text{①} \\ -2x + 7 < -1 & \text{②} \end{cases}$

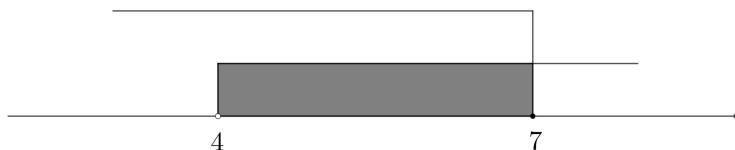
由①知 $3x \leq 21$

$x \leq 7$

由②知 $-2x < -8$

$x > 4$

$\therefore$ 原不等式组的解集为 $4 < x \leq 7$



19.(1) 解: $15 \times 60 = 900$ (秒)

$$l_1: s = 4t \quad (0 \leq t \leq 900)$$

$$\text{设 } l_2: s = 3t + b$$

$$\text{当 } t = 60 \times 2 = 120 \text{ 时, } s = 120 \times 3 + b = 0$$

$$\text{解得 } b = -360$$

$$\therefore l_2: s = 3t - 360 \quad (120 \leq t \leq 900)$$

(2) 解: $\Delta s = 4t - (3t - 360) = t + 360$

$$\text{当 } t = 900 \text{ 时, } \Delta s = 900 + 360 = 1260$$

$$1260 \div 400 = 3 \cdots 60$$

$\therefore$ 套圈 3 次

20.(1) 解: $\frac{40 - (-70)}{0.4} = 275 \text{ mV/mS}$

(2) 解: $\bar{x} = \frac{1}{5} \times [(-60) + (-65) + (-56) + (-68) + (-61)] = -62 \text{ mV}$

$$S^2 = \frac{1}{5} \times \{[(-60) - (-62)]^2 + [(-65) - (-62)]^2 + [(-56) - (-62)]^2 + [(-68) - (-62)]^2 + [(-61) - (-62)]^2\} = 17.2$$

(3) 解: $y = \begin{cases} 0, & 0 \leq x < 0.5 \\ 1, & x \geq 0.5 \end{cases}$

21.(1) 解: $\because |-15| - 10 \times 1 = 5$

$$\text{又 } \because 0 < 5 < 9$$

$$\therefore -15/ = 5$$

(2) 解: $m - 61 - 128 = 97 + 63$

$$/m - 1 - 8/ = /7 + 3/ /m - 9/ = 0$$

$$/m/ = 9$$

(3) 解: 错误, ① “ $/-7x + 70/ = 2$ ” 应为 “ $/-7x + 70/ = 1$ ”.

② “ $/-7x/ = 1$ ” 应为 “ $/-7x/ = 9$ ”.

由②得 $y = 8 - x$ ③.

$\because x, y$ 均为正整数,

$$\therefore \begin{cases} x > 0 \\ 8 - x > 0 \end{cases}$$

$$\text{解得 } 0 < x < 8.$$

由①得 $/2x + 9y/ = 3$ ④

把③代入④, 得 $/2x + 9(8 - x)/ = 3$

$$/2x + 72 - 9x/ = 3$$

$$/-7x + 72/ = 3$$

$$/-7x + 70/ = 1$$

$$\therefore -7x < 0$$

$$\therefore -7x/ = 9$$

规避: 优化代入方式, 使代数表达式尽可能保持非负, 系数为正

如何: ①先审视工具的前提, 理解透彻概念.

②抓住问题特殊性, 调整表达形式, 注重细节, 细节决定成败.

③最后才应用简化策略

22.(1) 解: $\because$ 四边形 $ABCD$ 为长方形

$$\therefore \angle BAE = \angle ABA' = 90^\circ$$

$\because$ 折叠

$$\therefore \angle BAE = \angle BA'E = 90^\circ$$

$$\therefore \angle AEA' = 360^\circ - \angle ABA' - \angle BAE - \angle BA'E = 90^\circ$$

$$\therefore \angle ABA' = \angle BA'E = \angle A'EA = \angle EAB = 90^\circ$$

$\therefore$ 四边形 $ABA'E$ 为矩形

$\because$ 折叠

$$\therefore BA = BA'$$

$\therefore$ 四边形 $ABA'E$ 为正方形

(2) 解: ① 如图, 四边形 $ABCD$ 为正方形, 四边形 $DEGH$ 为正方形, 四边形 $ECFG$ 为矩形, 且 $EC = 2, CF = 3$.

连接 AC, BD 交于点 I , 连接 EF, CG 交于点 J , 连接 AG .

$\because$ 四边形 $ABCD$ 为正方形, 四边形 $ECFG$ 为矩形

$\therefore I$ 为 CA 的中点, J 为 CG 的中点

$\therefore IJ$ 为 $\triangle CAG$ 的中位线

$$\therefore IJ \parallel \frac{1}{2}AG$$

$\because$ 四边形 $ECFG$ 为矩形

$$\therefore EG = CF = 3, CE = GF = 2$$

$\because$ 四边形 $DEGH$ 为正方形

$$\therefore DE = EG = GH = DH = 3, \angle DHG = 90^\circ = \angle HDE$$

$$\therefore DC = DE + EC = 5$$

$\because$ 四边形 $ABCD$ 为正方形

$$\therefore AD = DC = CB = BA = 5, \angle ADC = 90^\circ$$

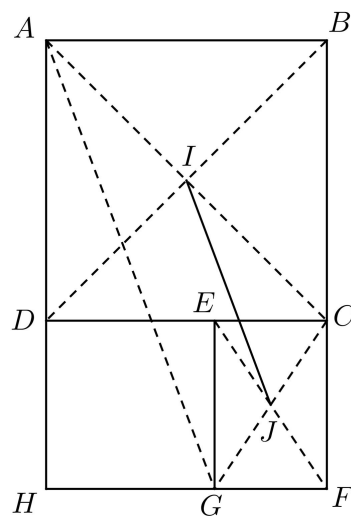
$$\therefore \angle ADC + \angle EDH = 180^\circ$$

$\therefore A, D, H$ 三点共线

$$\therefore AH = AD + DH = 8$$

$$\text{在 Rt}\triangle AHG \text{ 中, } AH = 8, HG = 3, AG = \sqrt{AH^2 + HG^2} = \sqrt{73}$$

$$\therefore IJ = \frac{1}{2}AG = \frac{\sqrt{73}}{2}$$



② 如图, 四边形 $ABCD$, 四边形 $EFDG$ 为正方形, 四边形 $HAFE$ 为矩形, 且 $HA = 2, HE = 3$. 连接 HF, AE 交于点 I , 连接 AC, BD 交于点 J , 连接 CE .

$\because$ 四边形 $HAFE$ 为矩形 $\therefore HA = EF = 2, HE = AF = 3$

$\because$ 四边形 $EFDG$ 为正方形 $\therefore EF = GD = EG = FD = 2, \angle EDF = 90^\circ = \angle GDF$

$$\therefore AD = AF + FD = 3 + 2 = 5$$

$\because$ 四边形 $ABCD$ 为正方形 $\therefore AD = DC = 5, \angle ADC = 90^\circ$

$$\therefore \angle ADC + \angle FDG = 180^\circ$$

$\therefore G, D, C$ 三点共线

$$\therefore GC = GD + DC = 5 + 2 = 7$$

$\because$ 四边形 $HAFE$ 为矩形, 四边形 $ABCD$ 为正方形

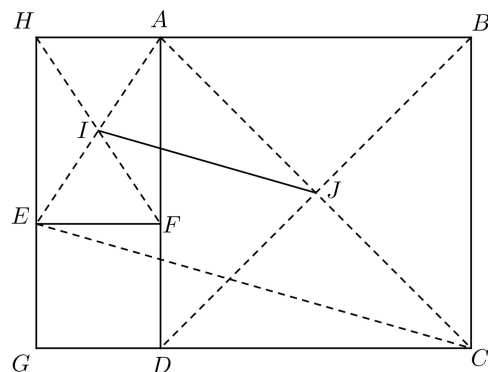
$\therefore I$ 为 EA 的中点, J 为 AC 的中点

$\therefore IJ$ 为 $\triangle AEC$ 的中位线

$$\therefore IJ \parallel \frac{1}{2}EC$$

$$\text{在 Rt}\triangle EGC \text{ 中, } EG = 2, GC = 7 \therefore EC = \sqrt{EG^2 + GC^2} = \sqrt{53}$$

$$\therefore IJ = \frac{1}{2}EC = \frac{\sqrt{53}}{2}$$



(3) 解: $18272 \div 384 = 47 \cdots 224$

$$384 \div 224 = 1 \cdots 160$$

$$224 \div 160 = 1 \cdots 64$$

$$160 \div 64 = 2 \cdots 32$$

$$64 \div 32 = 2$$

$$47 + 1 + 1 + 2 + 2 = 53 \text{ 个}$$

先将此长方形放大为原来的 $q_1 q_2$ 倍, 即长为 $p_1 q_2$, 宽为 $p_2 q_1$ 的一个长方形. 因为最终产生的正方形边长为长与宽的最大公因数, 所以此时最终得到正方形边长为 $\gcd(p_1 q_2, p_2 q_1)$. 再缩小为原来的 $q_1 q_2$ 倍, 可得由原长方形最终得到的正方形边长为 $\frac{\gcd(p_1 q_2, p_2 q_1)}{q_1 q_2}$.

23.(1)(i) 解: $\because x_1 = 1$

$$\therefore A(1, 0), OA = 1$$

$$\because OA = \frac{1}{4} OB$$

$$\therefore OB = 4$$

$\because B$ 在 x 轴正半轴上

$$\therefore B(4, 0)$$

$$\because a = 3, y \text{ 过点 } A(1, 0), B(4, 0)$$

$$\therefore y = 3(x - 1)(x - 4)$$

$$= 3x^2 - 15x + 12$$

(ii) 解: $3x^2 - 15x + 12 = mx + n$

$$3x^2 + (-15 - m)x + (12 - n) = 0 \quad (3)$$

$$\therefore x_3 + x_4 = -\frac{-15 - m}{3} = \frac{15 + m}{3} \quad (1)$$

$$\because CP = DP$$

$$\therefore \frac{x_3 + x_4}{2} = x_p = 3 \quad (2)$$

$$\therefore \text{由 } (1)(2) \text{ 可解得 } m = 3 \quad (4)$$

$$\text{当 } x = 3 \text{ 时, } y' = 3 \times 3 + n = p$$

$$\therefore n = p - 9 \quad (5)$$

$$\therefore \text{由 } (3)(4)(5) \text{ 知 } 3x^2 - 18x + (21 - p) = 0$$

$$\Delta = 18^2 - 4 \cdot 3(21 - p) > 0$$

$$\text{解得 } p > -6$$

$$\therefore y' = 3x + p - 9, p > -6$$

(2) 解: 设 $l: y = ix + j$, $l_0 \parallel l$, l_0 与 y 相切交于点 $E(x_5, y_5)$, 则可得 $l_0: y = ix + k$

$$ix + j = ax^2 + bx + c$$

$$\text{化简得 } ax^2 + (b - i)x + (c - j) = 0, x_1 + x_2 = -\frac{b - i}{a} \quad (6)$$

$$ix + k = ax^2 + bx + c$$

$$\text{化简得 } ax^2 + (b - i)x + (c - k) = 0, x_5 + x_5 = -\frac{b - i}{a} \quad (7)$$

$$\therefore \text{由 } (6)(7) \text{ 知 } x_5 = \frac{x_1 + x_2}{2}$$

$$y_3 - f\left(\frac{x_1 + x_2}{2}\right) = \frac{y_1 + y_2}{2} - f\left(\frac{x_1 + x_2}{2}\right)$$

$$= \frac{f(x_1) + f(x_2)}{2} - f\left(\frac{x_1 + x_2}{2}\right)$$

$$= \frac{1}{2} [(ax_1^2 + bx_1 + c) + (ax_2^2 + bx_2 + c)] - [a\left(\frac{x_1 + x_2}{2}\right)^2 + b\left(\frac{x_1 + x_2}{2}\right) + c]$$

$$= [a\left(\frac{x_1^2 + x_2^2}{2}\right) + b\left(\frac{x_1 + x_2}{2}\right) + c] - [a\left(\frac{x_1 + x_2}{2}\right)^2 + b\left(\frac{x_1 + x_2}{2}\right) + c]$$

$$= a\left[\frac{x_1^2 + x_2^2}{2} - \left(\frac{x_1 + x_2}{2}\right)^2\right]$$

$$= \frac{1}{4}a(x_1 - x_2)^2$$

$$\because x_1 \neq x_2$$

$$\therefore (x_1 - x_2)^2 > 0$$

在证明了 $x_5 = \frac{x_1+x_2}{2}$ 与 $y_3 - f(\frac{x_1+x_2}{2}) = \frac{1}{4}a(x_1 - x_2)^2$ 之后有以下两种证法.

法一: ① 当 $a > 0$ 时, $y_3 - f(\frac{x_1+x_2}{2}) > 0$, 由图知此时 l 与 y 必有两个交点

② 当 $a < 0$ 时, $y_3 - f(\frac{x_1+x_2}{2}) < 0$, 由图知此时 l 与 y 必有四个交点

法二: ① 假设 l 与 y 没有交点

$$\because y_3 = \frac{y_1 + y_2}{2}$$

$\therefore$ 由图知, y 不可能与 l 没有交点, 与假设矛盾.

② 假设 l 与 y 仅有一个交点

$$\because y_3 - f(\frac{x_1+x_2}{2}) = 0 = \frac{1}{4}a(x_1 - x_2)^2$$

$$\therefore x_1 = x_2$$

$\therefore$ 与 A, B 为两个不重合的点矛盾

综上, l 与 y 一定有两个交点

24.(1) 解: 连结 OD, OA

$$\because OF = OD = OA, DA = \sqrt{2}OF$$

$$\therefore DA = \sqrt{2}OF = \sqrt{2}OD = \sqrt{2}OA$$

$$\therefore DA^2 = OD^2 + OA^2$$

$$\therefore \angle DOA = 90^\circ$$

$$\therefore \angle ABD = \frac{1}{2}\angle DOA = 45^\circ$$

$$\because AB \parallel CD, AE \perp CD$$

$$\therefore AE \perp AB$$

$$\therefore \angle FAB = 90^\circ$$

$$\therefore \angle AHB = \angle DHE = 45^\circ$$

$\therefore$ 四边形 $DIKJ$ 为平行四边形

$$\therefore DI \parallel JK$$

$$\therefore \angle DHE = \angle HJK = 45^\circ$$

$$\therefore JL \perp AJ$$

$$\therefore \angle LJA = 90^\circ$$

$$\therefore \angle KJL = \angle LJA - \angle HJK = 45^\circ$$

$$\therefore JK = KL$$

$$\therefore \angle KJL = \angle KLJ = 45^\circ, \angle JKL = 90^\circ$$

$$\therefore CI \perp BD$$

$$\therefore \angle DIC = 90^\circ$$

$$\therefore AB \parallel CD$$

$$\therefore \angle ABD = \angle BDC = 45^\circ$$

$$\therefore \angle ICD = 45^\circ$$

在 $\triangle JKL$ 与 $\triangle DIC$ 中

$$\begin{cases} \angle KJL = \angle IDC = 45^\circ \\ JK = DI \end{cases}$$

$$\begin{cases} \angle JKL = \angle DIC = 90^\circ \end{cases}$$

$$\therefore \triangle JKL \cong \triangle DIC (ASA)$$

(2) 解: $\because CD \perp AF$

$$\therefore \angle DEJ = 90^\circ$$

$$\therefore \angle EDJ + \angle DJE = 90^\circ$$

$$\therefore \angle CDJ + \angle DJE + \angle HJK$$

$$+ \angle KJL = 180^\circ$$

$$\therefore \angle CDJ + \angle DJL = 180^\circ$$

$$\therefore JL \parallel DC$$

$\therefore$ 四边形 $JLCD$ 为平行四边形

$$\therefore LB \parallel JD$$

设 $\angle BCI = \alpha$, 则 $\angle IBC = 90^\circ - \alpha$

$\therefore A, B, C, D$ 四点共圆

$$\therefore \angle ABC + \angle ADC = 180^\circ$$

$$\therefore \angle ADC = 45^\circ + \alpha = \angle DCB$$

$$\therefore JD \parallel BL$$

$$\therefore \angle BCD = \angle JDC$$

$$\therefore \angle JDC = \angle CDA$$

$$\therefore 90^\circ - \angle JDC = 90^\circ - \angle CDA$$

$$\therefore \angle DJE = \angle DAE$$

$$\therefore JD = DA$$

$$\therefore \angle ABD = \angle BDC$$

$$\therefore DA = BC$$

$\therefore$ 四边形 $JDIK$ 为平行四边形

$$\therefore JD = KI$$

$\therefore$ 四边形 $JDCL$ 为平行四边形

$$\therefore JD = CL$$

连结 FB

$$\therefore \angle FAB = 90^\circ$$

$\therefore FB$ 为直径

$$\therefore FB = 2OF = OF + OB$$

$\therefore F, O, B$ 三点共线

连结 FL, FC

$\therefore FB$ 为直径

$$\therefore \angle FCB = 90^\circ$$

$$\therefore \angle FCD = \angle FCB - \angle DCB = 45^\circ - \alpha$$

$$\angle FCD = \angle DBF = 45^\circ - \alpha$$

$$\therefore \angle FBC = \angle DBC - \angle DBF = 45^\circ$$

$$\begin{aligned}
&\therefore \angle CFB = \angle CBF = 45^\circ \\
&\therefore CF = CB \\
&\therefore \angle FCL = 90^\circ, FC = CL \\
&\therefore \angle CFL = \angle CLF = 45^\circ \\
&\therefore \angle FLB = \angle FBL = 45^\circ, \angle LFB = 90^\circ \\
&\therefore \text{设 } FL = FB = 2x = 2OF, OF = x
\end{aligned}$$

连接 OL

在 $Rt \triangle OLF$ 中,

$$OL = \sqrt{OF^2 + FL^2} = \sqrt{5}x$$

在 $Rt \triangle FLB$ 中,

$$BL = \sqrt{FL^2 + FB^2} = 2\sqrt{2}x$$

$$\therefore KI = \frac{1}{2} \cdot 2KI = \frac{1}{2}(LC + CB)$$

$$= \frac{1}{2}BL = \sqrt{2}x$$

$$\therefore \frac{KI}{OL} = \frac{\sqrt{2}x}{\sqrt{5}x} = \frac{\sqrt{10}}{5}$$

(3) 解: $\because \angle AMB = 45^\circ$

$$\therefore \angle FHB = 135^\circ$$

$$\therefore \angle FHB + \angle FLB = 180^\circ$$

$\therefore F, H, B, L$ 四点共圆

$$\therefore \angle FHL = \angle FBL = 45^\circ$$

$$\therefore \angle JHL + \angle JLK + \angle LJH = 180^\circ$$

$\therefore L, K, H$ 三点共线

连结 LD, GL

$$\therefore \angle JLH = 45^\circ$$

$$= \angle FLJ + \angle FLD + \angle DLH$$

$$\angle FLC = \angle DLH + \angle HLC + \angle FLD = 45^\circ$$

$$\therefore \angle FLJ + \angle DLH = \angle DLH + \angle HLC = \angle DLC$$

为了证明 $\angle DLC = \angle CGL$,

有以下两种思路.

法一: 在 $\triangle GBC$ 与 $\triangle BDC$ 中

$$\therefore \begin{cases} \angle GCB = \angle DCB \\ \angle GBC = \angle BDC \end{cases}$$

$$\therefore \triangle GBC \sim \triangle BDC$$

$$\therefore \frac{BC}{CG} = \frac{DC}{BC}$$

$$\therefore BC = CL$$

$$\therefore \frac{CL}{CG} = \frac{DC}{CL}$$

在 $\triangle CLG$ 与 $\triangle CDL$ 中

$$\begin{cases} \frac{CL}{CG} = \frac{DC}{CL} \\ \angle LCG = \angle DCL \end{cases}$$

$$\therefore \triangle CLG \sim \triangle CDL$$

$$\therefore \angle CGL = \angle CLD$$

$$\therefore \angle CGL = \angle FLJ + \angle DLH$$

法二: $\because FL = FB$

$$\therefore \frac{FG}{FL} = \frac{FG}{FB}$$

$$\therefore DC \parallel AB$$

$$\therefore \frac{FG}{FB} = \frac{EF}{FA}$$

$$\therefore \frac{FG}{FL} = \frac{EF}{FA}$$

在 $\triangle DEH$ 与 $\triangle HJL$ 中

$$\therefore \begin{cases} \angle DEH = \angle HJL = 90^\circ \\ \angle EDH = \angle JHL = 45^\circ \end{cases}$$

$$\therefore \triangle DEH \sim \triangle HJL$$

$$\therefore \frac{DH}{HL} = \frac{EH}{JH}$$

连结 DF

$\because FB$ 为直径

$$\therefore \angle FDB = 90^\circ$$

$$\therefore \angle DFH = 45^\circ = \angle DHF$$

$$\therefore DF = DH$$

$$\therefore DC \perp AF$$

$$\therefore EF = EH$$

$$\therefore \frac{EH}{JH} = \frac{EF}{JH}$$

$$\therefore DJ = DA, DC \perp JA$$

$$\therefore JE = EA$$

$$\therefore JE - FE = EA - EH$$

$$\therefore JF = HA$$

$$\therefore JF + FH = HA + AF$$

$$\therefore JH = AF$$

$$\therefore \frac{EF}{JH} = \frac{EF}{AF}$$

$$\therefore \frac{DH}{HL} = \frac{EF}{AF}$$

$$\therefore \frac{FL}{FG} = \frac{AF}{DH}$$

$$\therefore \frac{FL}{FL} = \frac{HL}{HL}$$

在 $\triangle LFG$ 与 $\triangle LHD$ 中

$$\therefore \begin{cases} \frac{GF}{FL} = \frac{DH}{HL} \\ \angle LFB = \angle DHL = 90^\circ \end{cases}$$

$$\therefore \triangle LFG \sim \triangle LHD$$

$$\therefore \angle FGL = \angle HDL$$

$$\therefore \angle FGL = \angle FBL + \angle BLG$$

$$= 45^\circ + \angle BLG$$

$$= \angle HDL$$

$$= \angle HDC + \angle LDC$$

$$= 45^\circ + \angle LDC$$

$$\therefore \angle LDC = \angle BLG$$

可证 $\triangle CLG \sim \triangle CDL$

剩下思路与法一一致

